

*Your answer must be provided with descriptions how to get the answer.

1. (a) (4 points) Let $f(x) = \sin^{-1}(\sin x) + \cos^{-1}(\sin x) + \tan^{-1}\left(\frac{1}{x}\right)$.

Find $f'\left(\frac{1}{\sqrt{3}}\right)$.

2. (7 points) Evaluate $\lim_{x \rightarrow 0^+} (\coth x)^{\tanh x}$.

(b) (4 points) Use a linear approximation to estimate

$$\cos\left(\frac{\pi}{6} + 0.01\right) - \cos\left(\frac{\pi}{6}\right).$$

3. (7 points) If $f(x) = \int_0^{\sin^{-1}x} \sqrt{1+t^2} dt$ and $g(y) = \int_0^y f(y-t) dt$,
find $\left[\frac{d^2 g(y)}{dy^2} \right]_{y=1/2} = g''\left(\frac{1}{2}\right)$.

4. (7 points) Find the volume of the solid of revolution formed by rotating the finite region bounded by the graphs of $y = \sqrt{x-1}$ and $y = (x-1)^2$ about the y -axis.

5. (7 points) Evaluate the definite integral:

$$\int_0^1 \left(\sqrt{1-(x-1)^2} + \frac{1}{\sqrt{x} + \sqrt[3]{x}} \right) dx.$$

6. (9 points) Evaluate the indefinite integral:

$$\int \frac{\csc^2 x}{1 - \cot^3 x} dx.$$

1 (a) Let $f(x) = \sin^{-1}(\sin x) + \cos^{-1}(\sin x) + \tan^{-1}(\frac{1}{x})$. Find $f'(\frac{1}{\sqrt{3}})$

(b) Use a linear approximation to estimate

$$\cos(\frac{\pi}{8} + 0.01) - \cos \frac{\pi}{8}$$

(sol)

$$(a) f(x) = \sin^{-1}(\sin x) + \cos^{-1}(\sin x) + \tan^{-1}(\frac{1}{x})$$

$$\Rightarrow f'(x) = \frac{\cos x}{\sqrt{1 - \sin^2 x}} - \frac{\cos x}{\sqrt{1 - \sin^2 x}} + \frac{1}{1 + (\frac{1}{x})^2} \left(-\frac{2}{x^2}\right)$$

$$= \frac{-2x}{1 + x^2}$$

$$\Rightarrow f'(\frac{1}{\sqrt{3}}) = \frac{-\frac{2}{\sqrt{3}}}{1 + \frac{1}{3}} = -\frac{\sqrt{3}}{2}$$

$$(b) \text{ Let } f(x) = \cos x$$

$$f(\frac{\pi}{8}) = \cos \frac{\pi}{8}$$

$$\Rightarrow f'(x) = -\sin x$$

$$\Rightarrow f'(\frac{\pi}{8}) = -\sin \frac{\pi}{8} = -\frac{1}{2}$$

$\therefore$ The linearization of f at $\frac{\pi}{8}$ is

$$L(x) = f(\frac{\pi}{8}) + f'(\frac{\pi}{8})(x - \frac{\pi}{8})$$

$$= \cos \frac{\pi}{8} + (-\frac{1}{2})(x - \frac{\pi}{8})$$

and the linear approximation of f at $\frac{\pi}{8}$ is

$$f(x) \approx L(x) = \cos \frac{\pi}{8} - \frac{1}{2}(x - \frac{\pi}{8})$$

$$\therefore f(\frac{\pi}{8} + 0.01) \approx \cos \frac{\pi}{8} - \frac{1}{2}(\frac{\pi}{8} + 0.01 - \frac{\pi}{8})$$

$$\Rightarrow \cos(\frac{\pi}{8} + 0.01) - \cos \frac{\pi}{8} = -0.005$$

2. Evaluate $\lim_{x \rightarrow 0^+} (\cot x)^{\tan x}$

(sol) Let $y = (\cot x)^{\tan x}$

$$\Rightarrow \ln y = \tan x \ln(\cot x) = \frac{\ln(\cot x)}{\cot x}$$

$$\Rightarrow \lim_{x \rightarrow 0^+} \ln y = \lim_{x \rightarrow 0^+} \frac{\ln(\cot x)}{\cot x}$$

$$\stackrel{\text{L'Hosp.}}{=} \lim_{x \rightarrow 0^+} \frac{\frac{1}{\cot x} (-\csc^2 x)}{-\csc^2 x}$$

$$= \lim_{x \rightarrow 0^+} \frac{1}{\cot x}$$

$$= \lim_{x \rightarrow 0^+} \tan x$$

$$= 0$$

$$\therefore \lim_{x \rightarrow 0^+} (\cot x)^{\tan x} = \lim_{x \rightarrow 0^+} y$$

$$= \lim_{x \rightarrow 0^+} e^{\ln y}$$

$$= e^0$$

$$= 1$$

3. If $f(x) = \int_0^{x^{-1}} \sqrt{1+t^2} dt$ and $g(y) = \int_0^y f(y-x) dx$

find $\left[\frac{d^2 g(y)}{dy^2} \right]_{y=\frac{1}{2}} = g''\left(\frac{1}{2}\right)$

$$\begin{aligned}
 \text{(sol)} \quad g(y) &= \int_0^y f(y-x) dx \\
 &= \int_y^0 f(u) (-du) \\
 &= \int_0^y f(u) du
 \end{aligned}
 \quad \begin{cases} y-x = u \\ -dx = du \\ x=0 \rightarrow u=y \\ x=y \rightarrow u=0 \end{cases}$$

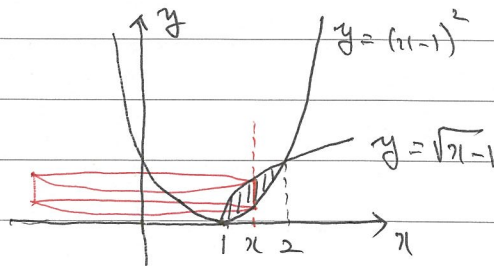
$$\Rightarrow g'(y) = f(y) = \int_0^{x^{-1}y} \sqrt{1+t^2} dt$$

$$\begin{aligned}
 \Rightarrow g''(y) &= f'(y) = \frac{d}{dy} \int_0^{x^{-1}y} \sqrt{1+t^2} dt \\
 &= \sqrt{1+(x^{-1}y)^2} \cdot \frac{1}{\sqrt{1-y^2}}
 \end{aligned}$$

$$\begin{aligned}
 \Rightarrow g''\left(\frac{1}{2}\right) &= \sqrt{1+(x^{-1}(\frac{1}{2}))^2} \cdot \frac{1}{\sqrt{1-\frac{1}{4}}} \\
 &= \sqrt{1+(\frac{\pi}{8})^2} \cdot \frac{2}{\sqrt{3}} \\
 &= \frac{\sqrt{3\pi^2+108}}{9}
 \end{aligned}$$

4. Find the volume of the solid of revolution formed by rotating the finite region bounded by the graphs of $y = \sqrt{x-1}$ and $y = (x-1)^2$ about the y -axis.

(sol)



The volume of the solid V is

$$V = \int_1^2 2\pi x (\sqrt{x-1} - (x-1)^2) dx$$

$$= 2\pi \left[\int_1^2 x \sqrt{x-1} dx - \int_1^2 x(x-1)^2 dx \right]$$

$$\textcircled{1} \int_1^2 x \sqrt{x-1} dx = \int_0^1 (t^2+1) t (2t) dt$$

$$\left(\begin{array}{ll} \sqrt{x-1} = t & x=1 \rightarrow t=0 \\ x-1 = t^2 & x=2 \rightarrow t=1 \\ dx = 2t dt \end{array} \right)$$

$$= 2 \int_0^1 (t^4 + t^2) dt$$

$$= \frac{16}{15}$$

$$\textcircled{2} \int_1^2 x(x-1)^2 dx = \int_1^2 (x^3 - 2x^2 + x) dx$$

$$= \frac{7}{12}$$

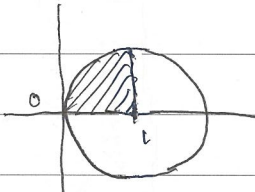
$$= 2\pi \left(\frac{16}{15} - \frac{7}{12} \right)$$

$$= \frac{29}{30} \pi$$

5. Evaluate the definite integral:

$$\int_0^1 \left(\sqrt{1-(x-1)^2} + \frac{1}{\sqrt{x} + \sqrt[3]{x}} \right) dx$$

(sol) ① $\int_0^1 \sqrt{1-(x-1)^2} dx = \frac{\pi}{4}$

$$\left(\begin{array}{l} \therefore \text{let } y = \sqrt{1-(x-1)^2} \\ \Rightarrow (x-1)^2 + y^2 = 1, y \geq 0 \end{array} \right)$$


② $\int_0^1 \frac{1}{\sqrt{x} + \sqrt[3]{x}} dx = \int_0^1 \frac{6x^{\frac{5}{6}}}{x^3 + x^2} dx$

$$\left(\begin{array}{l} \sqrt[6]{x} = t \\ x = t^6 \\ dx = 6t^5 dt \\ x=0 \rightarrow t=0 \\ x=1 \rightarrow t=1 \end{array} \right)$$

$$= \int_0^1 \frac{6t^3}{t+1} dt$$

$$= 6 \int_0^1 \left(t^2 - t + 1 - \frac{1}{t+1} \right) dt$$

$$= 6 \left[\frac{1}{3} t^3 - \frac{1}{2} t^2 + t - \ln|t+1| \right]_0^1$$

$$= 6 \left(\frac{1}{3} - \frac{1}{2} + 1 - \ln 2 \right)$$

$$= 5 - 6 \ln 2$$

$$\therefore \int_0^1 \left(\sqrt{1-(x-1)^2} + \frac{1}{\sqrt{x} + \sqrt[3]{x}} \right) dx = \frac{\pi}{4} + 5 - 6 \ln 2$$

6. Evaluate the infinite integral:

$$\int \frac{\csc^2 x}{1 - \cot^3 x} dx$$

$$(sol) \int \frac{\csc^2 x}{1 - \cot^3 x} dx = \int \frac{-1}{1 - t^3} dt$$

$$\left(\begin{array}{l} \cot x = t \\ -\csc^2 x dx = dt \end{array} \right) = \int \frac{1}{(t-1)(t^2+t+1)} dt$$

$$\begin{aligned} \therefore \frac{1}{(t-1)(t^2+t+1)} &= \frac{a}{t-1} + \frac{bx+c}{t^2+t+1} \\ &= \frac{(a+b)t^2 + (a-b+c)t + a-c}{(t-1)(t^2+t+1)} \\ \left\{ \begin{array}{l} a+b=0 \\ a-b+c=0 \\ a-c=1 \end{array} \right\} &\Rightarrow \left\{ \begin{array}{l} a=\frac{1}{3} \\ b=-\frac{1}{3} \\ c=-\frac{2}{3} \end{array} \right. \end{aligned}$$

$$\begin{aligned} &= \frac{1}{3} \int \left(\frac{1}{t-1} - \frac{t+2}{t^2+t+1} \right) dt \\ &= \frac{1}{3} \int \left(\frac{1}{t-1} - \frac{\frac{1}{2}(2t+1) + \frac{3}{2}}{t^2+t+1} \right) dt \\ &= \frac{1}{3} \int \left(\frac{1}{t-1} - \frac{1}{2} \frac{2t+1}{t^2+t+1} - \frac{3}{2} \frac{1}{(t+\frac{1}{2})^2 + (\frac{\sqrt{3}}{2})^2} \right) dt \\ &= \frac{1}{3} \ln|t-1| - \frac{1}{6} \ln(t^2+t+1) \\ &\quad - \frac{1}{3} \cdot \frac{2}{\sqrt{3}} \tan^{-1} \left(\frac{t+\frac{1}{2}}{\frac{\sqrt{3}}{2}} \right) + C \\ &= \frac{1}{3} \ln|\cot x - 1| - \frac{1}{6} \ln(\cot^2 x + \cot x + 1) \\ &\quad - \frac{1}{\sqrt{3}} \tan^{-1} \left(\frac{2\cot x + 1}{\sqrt{3}} \right) + C \end{aligned}$$